Input and protection

File: input.kicad_sch

Power channels

File: channels.kicad_sch

Control

File: control.kicad_sch

Power supplies

File: power.kicad_sch

**4-channel electronic load — 60 V / 20 A / 300 W**

Each channel: 0–5 A, max 75 W per MOSFET. Above 15 V the limit is power (300 W / V), enforced by firmware + INA228 ALERT.

Setpoint: DAC8565 0–2.5 V –> 25:1 divider –> 0–100 mV on 20 mOhm shunt = 0–5 A. The divider is referenced to the cold Kelvin pad of the shunt (SHNx).

ARM (active high, 10k pull-down) enables the load. LM393 (OVP 64.7 V, OTP ~100 C) and the INA228 ALERTs pull it low in hardware.

ARM low –> CLAMP high –> clamp FETs short VSET1..4 to ground. MCU in reset = load off.

Single GND = DUT– (star point at stud J2). Control over WiFi only; the Wemos USB is NOT isolated: use it only for programming with the DUT disconnected.

ESP8266 firmware: after SPI.begin() re-assign GPIO12 as output (ARM_DRV); GPIO16 = ARM_FB; A0 = NTC1 (D1 mini internal 220k/100k divider).

Front panel (rev 0.3): PCF8574 @0x20 (encoder A/B/SW, OUTPUT button, LED), INT on GPIO0; OLED @0x3C on the same bus. OUTPUT ON enables ARM, never automatic.

**Exceptions to the SMD SOIC/SOP + 1206 passives rule**

TO-247 power MOSFETs (mounted on the spreader), WSK2512 Kelvin shunts, D2PAK reverse-polarity diode, SMC/SMA TVS,

INA228 in VSSOP-10 (0.5 mm pitch), DAC8565 in TSSOP-16, AP63205 in TSOT-23-6, SRN4018 inductor, THT connectors.

**To verify on the bench**

Wemos antenna off the board edge and away from the cooler (or external antenna on U.FL); CC loop stability analysed (C 1n C0G / R 1k / Rg 100R, rev 0.2), to confirm on the bench; D1 I2t vs F1; IXTH75N10L2 FBSOA at 60 V and Tc 90 C.

Hierarchical schematic: one sheet per functional block

**IV3HGA**

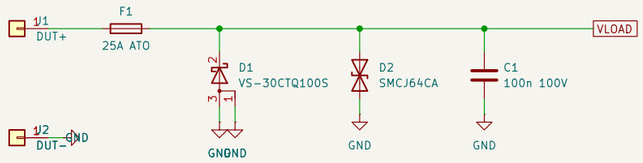
Sheet: /

File: eload_4ch.kicad_sch

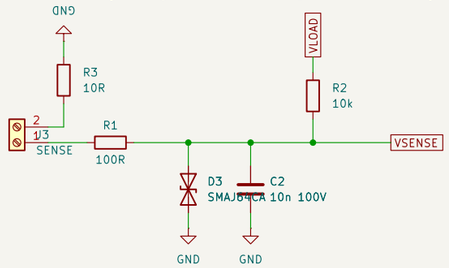
***Title: Electronic Load 4ch 60V 20A 300W — Index***

| Size: A3 | Date: 2026-10-25 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/5 |

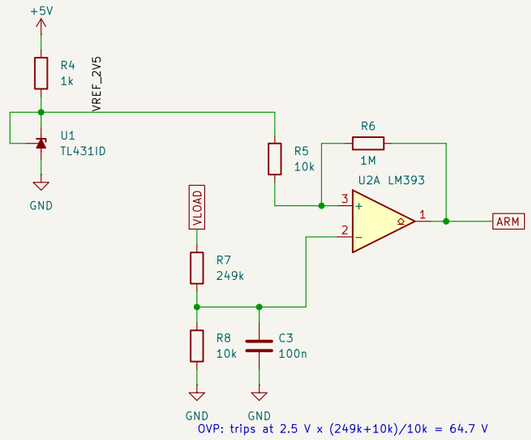
## DUT terminals and power-path protection



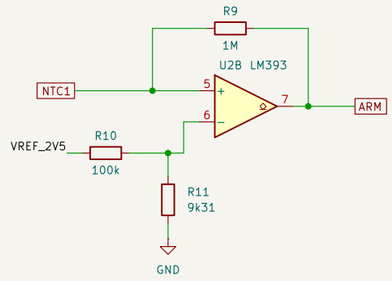

## Remote sense (voltage measured by the INA228s)




## 2.5 V reference and OVP / OTP comparators (open-collector outputs on ARM)

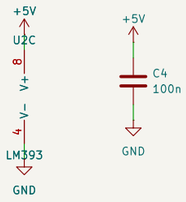



OVP: trips at 2.5 V x (249k+10k)/10k = 64.7 V




OTP: NTC 10k B3950 at -100 C = 0.21 V < OTP_REF

## Comparator supply




DUT terminals, fuse, reverse-polarity diode, TVS, remote sense, hardware OVP/OTP

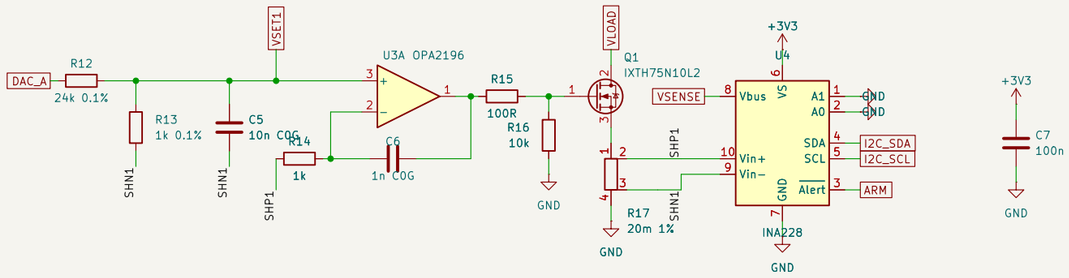
**IV3HGA**

Sheet: /Input and protection/
File: input.kicad_sch

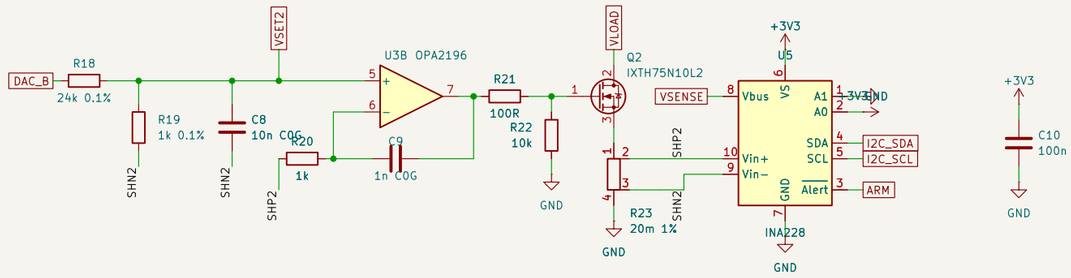
***Title: Electronic Load 4ch 60V 20A 300W — Input and protection***

| Size: A3 | Date: 2026-10-25 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 2/5 |

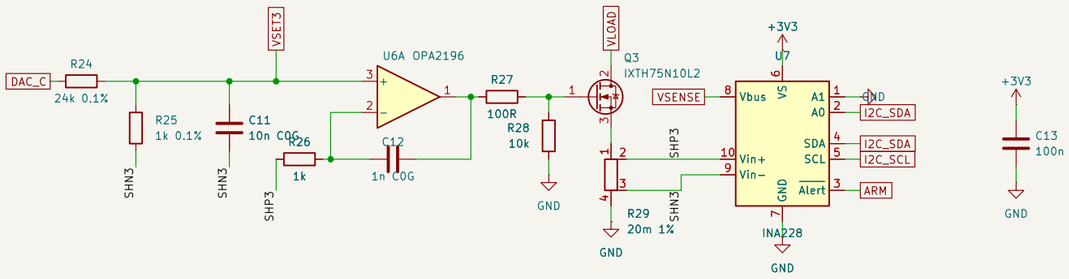
## Channel 1 — 0–5 A, 75 W — INA228 @0x40



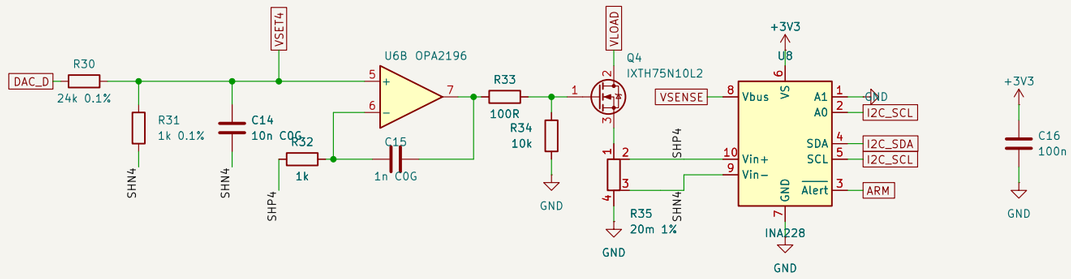

## Channel 2 — 0–5 A, 75 W — INA228 @0x41



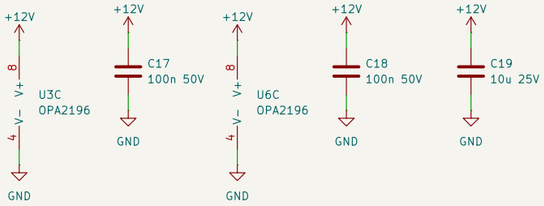

## Channel 3 — 0–5 A, 75 W — INA228 @0x42




## Channel 4 — 0–5 A, 75 W — INA228 @0x43




## Op-amp supply




4 CC channels: setpoint, OPA2196 loop, linear MOSFET, Kelvin shunt, INA228

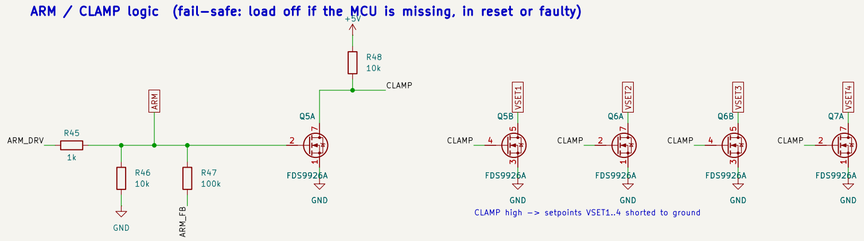
**IV3HGA**

Sheet: /Power channels/
File: channels.kicad_sch

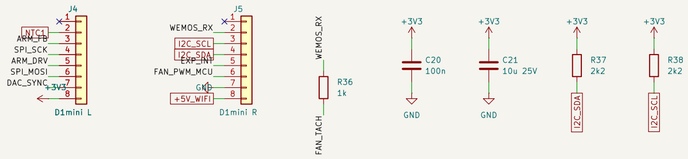
***Title: Electronic Load 4ch 60V 20A 300W — Power channels***

| Size: A3 | Date: 2026-10-25 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/5 |

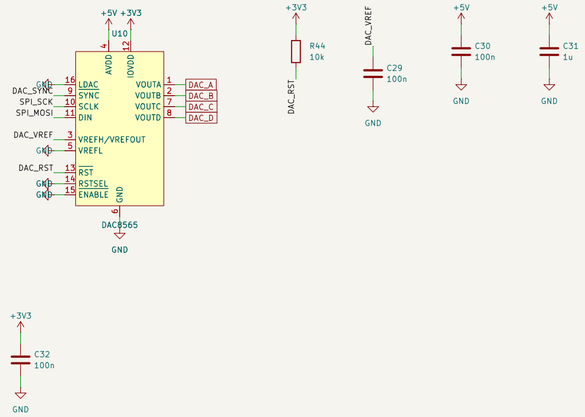
## ARM / CLAMP logic  (fail-safe: load off if the MCU is missing, in reset or faulty)



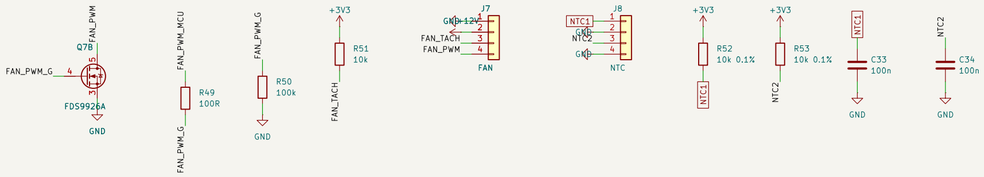

## WiFi module Wemos D1 mini Pro (ESP8266) on socket strips



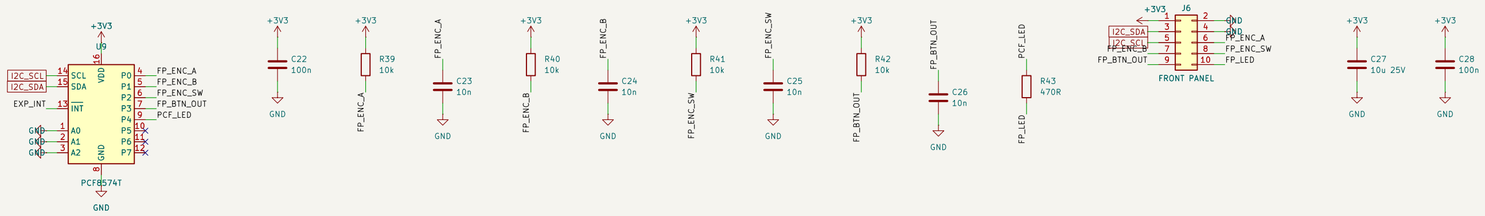

## Quad 16-bit DAC (internal 2.5 V reference)




## 4-wire PWM fan and NTC sensors on the spreader




## Front panel: I2C OLED, push encoder, OUTPUT button (rev 0.3)







Wemos D1 mini Pro (ESP8266), DAC8565, ARM/CLAMP logic, fan, NTC, front panel

**IV3HGA**

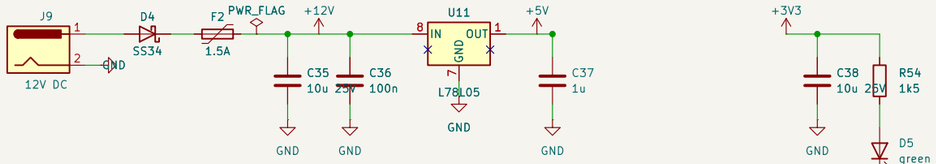
Sheet: /Control/
File: control.kicad_sch

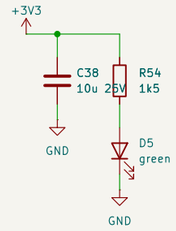
***Title: Electronic Load 4ch 60V 20A 300W — Control***

| Size: A2 | Date: 2026-10-25 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/5 |

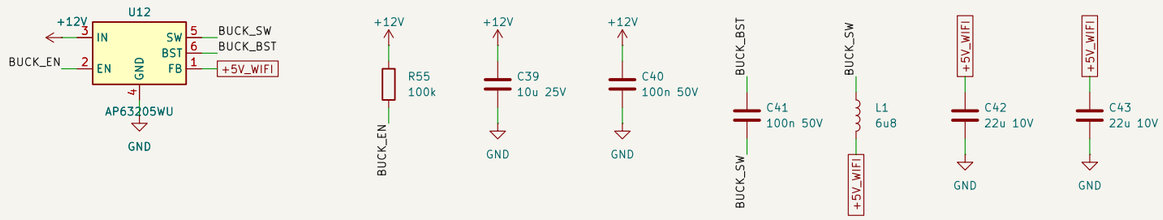
## 12 V auxiliary input (dedicated isolated adaptor) and regulators

+3V3 supplied by the Wemos module regulator (3V3 pin)




## Buck 12 V –> 5 V (AP63205) for the Wemos only: analog 5 V stays on the 78L05










12 V auxiliary input, analog 5 V (78L05), 5 V buck for WiFi

**IV3HGA**

Sheet: /Power supplies/
File: power.kicad_sch

***Title: Electronic Load 4ch 60V 20A 300W — Power supplies***

| Size: A3 | Date: 2026-10-25 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/5 |